Mudiam Hemanth Reddy

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Profile

B.Tech Artificial Intelligence & Data Science student with hands-on experience building software applications, machine learning systems, and data-driven projects. Skilled in Java, Python, JavaScript, frontend development, and AI/ML libraries. Experienced in developing end-to-end applications involving machine learning APIs, real-time data processing, and interactive dashboards. Interested in software engineering, artificial intelligence, and building practical solutions to real-world problems.

Education

B.Tech, Artificial Intelligence & Data Science — Amrita Vishwa Vidyapeetham, Aug 2024 – May 2028 | CGPA: 8.26/10

Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Machine Learning, Deep Learning

Intermediate (MPC), Class 12 — Narayana, Telangana — 94.2%

Schooling — Sri Chaitanya School, Meerpet, Telangana (State Board)

Experience

AI & Data Science Intern — XYlofy AI (Remote)

- Built an end-to-end real-time fraud detection system for identifying suspicious financial transactions using machine learning and explainable AI.
- Evaluated LightGBM, XGBoost, and Isolation Forest, and optimized classification thresholds for fraud detection.
- Applied SHAP to explain model predictions and developed an interactive Streamlit dashboard for fraud-risk analysis.

Technologies: Python, Pandas, Scikit-learn, LightGBM, XGBoost, SMOTE, SHAP, Streamlit

Technical Skills

Languages: Java, Python, JavaScript, C++, Scala, HTML, CSS
AI/ML: Scikit-learn, TensorFlow, Pandas **Backend:** FastAPI, REST APIs
Data/Tools: Apache Spark, Hadoop HDFS, Kafka, Git, Docker

Projects & Research

AI Cyberattack Detection System (SDN-OSHLD)

- Built a network intrusion detection system using machine learning models on the CICIDS2017 benchmark.
- Achieved 99.2% classification accuracy across three network datasets and deployed the model through a FastAPI REST API with sub-100 ms response time during testing.

Technologies: Python, Scikit-learn, Pandas, FastAPI, Docker Link: GitHub

Real-Time Taxi Demand, Fare & ETA Prediction

- Built a big-data pipeline for taxi demand, fare, and trip-duration prediction using Hadoop and Apache Spark, as part of a four-person team.
- Processed 17+ GB of trip data and 11M+ records using Spark.
- Integrated weather and geographic zone data, and contributed to ML pipelines for hourly demand, fare, and ETA prediction using Spark MLlib.
- Added Kafka-based streaming for near-real-time data processing.

Technologies: Hadoop HDFS, Apache Spark, Scala, Spark MLlib, Kafka Link: GitHub

Adaptive Graph Engine — GPU-Accelerated Graph Processing

- Built a graph processing engine in C++ using CUDA to run traversal algorithms (BFS, PageRank, shortest-path) on the GPU for faster performance on large graphs.
- Added a thread-pool scheduler to manage parallel workloads and recover from simulated worker failures.
- Connected the C++ engine to a Python/FastAPI backend via PyBind11 and built a simple dashboard to visualize live performance metrics.

Technologies: C++, CUDA, FastAPI, PyBind11 Link: GitHub

Electric Vehicle Stability Improvement via NMPC (TVC + RWS)

- Developed an independent MATLAB/Simulink framework for Nonlinear Model Predictive Control (NMPC) combining Torque Vectoring Control (TVC) and Rear-Wheel Steering (RWS) to enhance EV lateral stability and handling.
- Integrated a 2-DOF vehicle dynamics model with non-linear Fiala tire mechanics, dynamic axle load distribution, and a reference yaw-rate generator.
- Implemented a Pontryagin's Minimum Principle (PMP) solver with soft sideslip constraint relaxation for low-latency performance, benchmarked against a direct interior-point NLP solver across Double Lane Change and Fishhook maneuvers.

Technologies: MATLAB, Simulink, Nonlinear MPC, Vehicle Dynamics Modeling Link: GitHub

Certifications

AI & Data Analytics Internship Certificate — XYlofy AI, 2026
Getting Started with AI on Jetson Nano — NVIDIA Deep Learning Institute
GenAI Powered Data Analytics Job Simulation — 2026

Activities & Hobbies

National-level hackathon participant and team lead.